Swinburne University of Technology Sarawak
Faculty of Engineering, Computing and Science

School of Information and Communications Technologies

COS10005 Web Development

Assignment 2 - *Semester 2, 2026*

Late submission penalty is 5% of the total available marks per day
This is an Individual assignment.

Contribution to Final Assessment: 40%

Purpose of the assignment

In this assignment you will extend your Assignment 1 website by adding a link/button/icon to each service offered by your Website. This link would ease users to send an inquiry regarding a certain service. All inquiry information must be validated before it can be submitted successfully.

In short you will:

- Use *client-side storage* to transfer data between pages and within a page.
- Use only *JavaScript* to validate forms and provide user enquiry.

Section 1: Essential specified requirements

1. Data Transfer between Web pages

Add a "Subject" input field into your existing **enquiry.html** page. This page must be accessible from any page where a product/service is displayed. When a link/button on a specific product is clicked, the user will be redirected to the **enquiry.html** page and the "Subject" input field will be pre-filled with the relevant product name/description. You will need to use JavaScript to obtain this data and store it locally using either **session objects** or **cookies**. You can decide the format of the "Subject" input field's value, such as `RE: Enquiry on [servicename]'.

If you wish to make other alterations to your Assignment 1, that is OK. All newly added pages in the website should be directly or indirectly accessible from your main website menu.

Remember to implement your web pages using the standard HTML5 elements (*unless indicated otherwise*) that are also well-formed XML.

2. Data transfer in the same page

- Add a "Subject" input field into your existing **enquiry.html** page. This page must be accessible from *Explorer pages*. For example, when a link/button on a specific plant is clicked, the user will be redirected to the **enquiry.html** page and the "Subject" input field will be pre-filled with the relevant subject. You will need to use JavaScript to obtain this data and store it locally using either **session objects** or **cookies**. You can decide the format of the "Subject" input field's value, such as `RE: Enquiry on [Families/Genera/Species]'.

3. Data validation and input checking of Forms

You will demonstrate different techniques for using client-side validation of data entered into the form. On the enquiry.html page, use JavaScript for input validation. Replace HTML5-specific input types that have embedded validation functionalities (e.g. *email*, *number* and *tel*) types with the normal HTML input *text* type.

4. Populate drop-down list using Javascript

Create an array to store the product list in Javascript. Change the drop-down list on the following:

- enquiry.html to obtain the list of products from the array. You are to programmatically create a new DOM node for each product and append it to the HTML element.
- Work on the navigation bar.

Section 2: UI Testing (Report)

UI testing is critical in web application development. Instead of having testers click through the application to verify data and action flows visually (manual testing), automated testing reduces costs and ensure actionable results whereby test scripts are written for each test case. In this task, perform a survey on the available tools used for automating Web UI testing. Write a report to describe on any TWO(2) of the available web UI testing tools that could be adapted in your assignment, and compare their advantages and disadvantages.

Enhancements using JavaScript

You should complete the above requirements before attempting enhancements.

Similar to Assignment 1, you have the opportunity to implement enhancements to your Website using techniques not covered in the tutorials. Each enhancement must be listed of a page called **enhancement2.html**. The entries on this page should:

- Briefly describe the interaction required to trigger the event and what a programmer has todo to implement the feature.
- Provide a hyperlink to the page where the enhancement is implemented in your website.
- Reference any 3rd party contribution to the enhancement

The JavaScript enhances features should be in a separate **enhancements.js** file. Make sure there are adequate comments to explain the enhancement (including the source for its code, if applicable). ***Examples*** of JavaScript and other enhancements that you might include:

- Pop-up a preview page prior feedback formsubmission.
- Use **regular expression** patterns in JavaScript to validate the data in more depth. In relationto this, you may introduce additional validation criteria when validating the form.
- The "Navigation Menu" automatically highlights the current page being viewed.
- Create an extra client side JavaScript dynamic effect: e.g. slideshow, random image displayed onload, etc. The code and structure of this is open, but must be documented and explained as clearly as possible.
- Use the HTML Form Element to manipulate/validate the form (e.g. see https://developer.mozilla.org/en-US/docs/Web/API/HTMLFormElement)
- Etc.

Any extensions that are not listed and linked on the page enhancement2.html will not be assessed. A maximum of 5 extensions will be assessed.

Web Site Folder Structure and Deployment Requirements

Your website folder structure should follow the same structure as in the previous assignment
All links to your files should be relative. Do not use absolute links, as these links will probably be broken when files are transferred for marking. **No marks will be allocated if links are broken**.

Assignment Submission

An electronic copy of your assignment should be submitted through Canvas on or before the due date.

- Make sure all your files are in the correct folders and compress the entire folder into a zip filenamed "**yourname_assign2.zip**". You are to include your report as well.